Exno:1

Aim:- Study of various network commands used in Linux & windows

Basic Networking Commands

1). arp -a : IP address of your computer

~~O/P:-~~ O/P:- -gateway (10.0.2.2) at 52:55:0a:00:02 [ether] on enf083

2). hostname: Displays name of your computer

O/P:- fedora

3). ipconfig/all: Displays complete IP configuration

O/P:- Windows IP configuration

Host Name . . . . . : DESKTOP-AQOHL8O

Physical Address . . . : f8-ED-FC-5C-06-39

IPv4 Address . . . : 172.16.10.163 (Preferred)

(w) Subnet Mask . . . . : 255.255.252.0

4) nbtstat -a; Displays NetBIOS information of a remote computer

O/P:- NBTSTAT [[-a RemoteName] [-A IP Address] [-c] [-n]
[-r] [-R] [-RR] [-s] [-S] [interval]]

(w) 5). netstat : Display active network connections and ports

O/P:- Active Connections (w/o servers)

Proto	Local Address	Foreign Address	State
udp	fedora:bootpc	gateway:bootps	ESTABLISHED

6). nslookup: Finds the IP address of a website

I/P:- nslookup google.com

O/P:- Server: 127.0.0.53

Address: 127.0.0.53#53

Non-authoritative answer:-

Name: google.com

Address: 142.251.221.206

(W)
7). pathping: Shows the path and packet loss to a destination

I/P:- pathping google.com

O/P:- Tracing route to google.com [142.251.221.206]
over a maximum of 30 hops:
0 DESKTOP-AQOHL80 [172.16.10.163]
1 172.16.8.1
2 137.97.163.129

8). ping:- Checks whether another device is reachable

I/P:- ping google.com

O/P:- 64 bytes from Pnmaa-da-in-f14.1e100.net (142.251.221.206):
icmp-seq=1 ttl=255 time 6.65ms
⋮
---google.com ping statistics---
19 packets transmitted, 18 received 5.263161. packetloss,
time 18069ms

(W)
9). Route: Displays the routing table

I/P:- route print

O/P:- Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.16.8.1	172.16.10.163	281
127.0.0.0	255.0.0.0	On-Link	127.0.0.1	331
⋮				

10). ip: ip <options> <object> <command>

a). To show IP:- ip address show

O/P:- 1: lo:
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever

b). To assign IP:- ip address add 192.168.1.254/24 dev enp0s3

O/P:- 2: enp0s3:
inet 192.168.1.254/24 scope global enp0s3
valid_lft forever preferred_lft forever

c). To delete an IP: ip address del 192.168.1.254/24 dev enp0s3

O/P:- 2: enp0s3:

inet 10.0.0.15/24 brd 10.0.2.255 scope global dynamic

valid_lft forever preferred_lft forever no prefix route enp0s3

d)i) Alter status of interface: ip link set eth0 up

ii) " " down: ip link set eth0 down

iii). By enabling promiscuous mode: ip link set eth0 promisc on

O/P:- for UP,

enp0s3: mtu 1500 qdisc fq_codel state UP group default qlen 1000

For DOWN,

enp0s3: mtu 1500 qdisc fq_codel state DOWN group default qlen 1000

For promiscuon, 2: enp0s3 <BROADCAST, PROMISC)

altname enx080027d7ec87

(No inet) (removed)

e)i) Add default route via gateway: ip route add default via 192.168.1.0/24 dev enp0s

ii). O/P:-

eth0: mtu 1500 qdisc fq_codel state UP group

default qlen 1000

f)i). Add route to default via gateway: ip route add 192.168.1.0/24 via 192.168.1.254

O/P:-

ii). Add route to default on device: ip routeadd 192.168.1.0/24 dev enp053

O/P:-

iii). Delete route via gateway: ip route delete 192.168.1.0/24 via 192.168.1.254

O/P:-

iv) Display the route taken; ip route get 10.10.1.4

O/P:-

10.10.1.4 via 192.168.1.254 dev eth0 src 10.0.2.15

cache

12). ifconfig: Configuring & troubleshooting network

O/P:-

eth0: flags=4163<UP, BROADCAST, RUNNING, MULTICAST]

mtu 1500 inet 10.6.2.15

lo: flags = 73<UP, LOOPBACK, RUNNING> mtu

65536 inet 127.0.0.1

12) mtr: mtr < options > hostname IP

a). Shows statistics with time & loss%.

mtr google.com

O/P:- Host Loss% Snt Last Avg Best Wrst StDev

1. prnaa-au-in-f14.1e100.net 0.0% 68 9.0 18.2 3.2 60.2 10.7

b). show numeric IP address

mtr -g google.com

O/P:- google.com (142.251.221.174)

c). shao numeric IP address & hostname

mtr -b google.com

O/P:- 142.251.221.174, prnaa-qu-in-f14.1e100.net

d). set numbers of pings

mtr -c 10 google.com

O/P:- Host Loss% Snt Last Avg Best Wrst StDev

prnaa-au-in-f14.1e100.net 0.0% 69 9.0 17.2 3.2 60.1 10.7

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13) tcpdump

i). Install: dnf install -y tcpdump

ii). tcpdump -D

iii). tcpdump -i eth0 : Capture traffic on eth0

09:25:30.619674 IP mirrors.twds.com.tw.http > fedora.36222: Flags [.],

O/P:- win 65535, length 1440

485192 packets captured | 519028 packets received by filter | 33705 packets dropped

iv) tcpdump -i eth0 -c 10 : Capture traffic to and from one host

O/P:- 10 packets captured

20 packets received by filter

0 packets dropped by kernel

v). To find traffic coming from & going to 8.8.8.8

tcpdump -i eth0 -c 10 host 8.8.8.8

O/P:- 11 packets captured

21 packets received by filter

0 packets dropped by kernel

vi). Traffic coming 8.8.8.8

tcpdump -i eth0 src host 8.8.8.8

O/P:- 11 packets captured

20 packets received by filter

1 packets dropped by kernel

To & from

- Host

viii). Outbound traffic going to 8.8.8.8
tcpdump -i eth0 dst host 8.8.8.8
O/P 11 packets captured
21 packets received by filter
0 packets dropped by kernel

viii) Traffic from network
tcpdump -i eth0 not 10.1.0.0 mask 255.255.255.0
O/P 10 packets captured
21 packets received by filter
0 packets dropped by kernel

ix). tcpdump -i eth0 net 10.1.0.0/24
O/P 10 packets captured
20 packets received by filter
0 packets dropped by kernel

x). DNS port 53 traffic
tcpdump -i eth0 port 53
O/P 10 packet captured
20 packets by filter
0 packets by kernel

xi). Specific host, tcpdump -i eth0 host 8.8.8.8 and port 53
O/P 10 packets captured
20 packets by filter
0 packets by kernel

xii). HTTPS traffic
tcpdump -i eth0 -c 10 host www.google.com and port 443
O/P 10 packets captured
20 packets by filter
0 packets by kernel

xiii). All port except port 80 and 25,
tcpdump -i eth0 port not 53 and not 25
O/P 10 packets captured
20 packets by filter
0 packets by kernel

To & from -> host

To & from -> network

To & from -> port

4) Ping :- It's a tool to verify connectivity to TCP/IP by sending ICMP.

i). ping google.com
O/P 64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174):
icmp_seq=1 ttl=56 time=10.7ms

ii). ping -c 10 google.com : to ping a host 10 times
O/P 64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174): icmp_seq=1 ttl=56 time=10.6ms
=10.4ms

Returns ping 10 times

15). nmcli : reconfig ethernet connection

i). nmcli connection show

O/P:- NAME UUID Type

Wired connection 1 d7d7305b-c2d9-3cb3-ab4b ethernet

lo 989a3b51-9393-466a-bebb loopback

ii). nmcli connection add con-name <connection-name> ifname <device-name> type ethernet (Syntax)

To rename a connection profile :-

iii). nmcli connection modify "Wired connection 1" : rename connection

iv). nmcli connection show : display setting of connection

O/P:- ipv4.method : auto

v). nmcli connection modify "wiredconne 1" ipv4.method auto : config IPv4

O/P:- Modify the IP address

vi). nmcli connection modify "wiredconne 1" ipv6.methodauto : config IPv6

O/P:- Modify the IP address

vii). nmcli connection up Internal-LAN :

Verification

O/P:- Connection successfully activated

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16). Display DNS:-

i). cat : /etc/resolv.conf

O/P:-

ii). ping <host-name or IP-address> : to verify host is to send packets to other host

O/P:- default via 192.168.1.254 dev eth0

iii). ip -6 route show :- Shows IPv6 default

O/P:- fe80::625c:f0ff:fe37:2a1:164 dev eth0 proto ra metric

default via 192.168.1.254 dev eth0

iv). ip route show

O/P:- default via 192.168.1.254 dev eth0

Student Observation:-

1). Which command is used to find the reachability of a host machine from your device?

Ans Ping Syntax: ping <hostname>

2). Which command will be give the details of hops taken by a packet to reach its destination?

Ans traceroute

3). Which commands displays the ip configuration of your machine?

Ans ipconfig or ifconfig

4). Which command displays the TCP port status in your machine?

Ans netstat

5). Write the command to modify the IP configuration in a linux machine

Ans ip address add 192.168.1.254 /24 dev enp0s3